

IMPROVEMENT CHANGELOG

One row per experiment. Evidence cells point at a run directory, re-scorable at any time, and every run backing a claim is promoted into `runs-archive/` where `npm run verify` reproduces it without credentials.

STAGE	WHAT YOU TRIED AND WHY	EVIDENCE	DECISION / LEARNING
Untreated	Scored the pristine fixtures to establish the floor and prove every seeded defect registers	<code>runs-archive/before</code> , 0/12 pass, axe counts 0 to 9 per case, keyboard failures on 03, 06, 07, 09, 12	Starting point established. Cases 03, 06, 07, 09 fail with near-zero axe counts, the keyboard gate measures what the linter cannot
Baseline, benchmark v1	One direct structured-output call per case against the original spec-style contract	<code>runs/baseline-2026-08-30T15-26-59</code> , 12/12 pass, \$0.62 total, modal and dropdown probes both green	Benchmark saturated. The contract itemized every requirement (focus trap recipe, ARIA listbox pattern, click parity), so a one-shot call just implemented the spec. The eval was measuring spec-following, not defect discovery
Agent v1, benchmark v1	<code>npm run agent -- --variant v1-notools</code> against the same spec-style contract, run to complete the saturation picture	<code>runs/agent-v1-notools-2026-08-30T15-36-10</code> , 12/12 pass, \$1.70 total, mean 8.4 turns per case	Confirmed the benchmark cannot discriminate. Iteration without tools matched the one-shot result at almost triple the cost, so every arm saturates when the contract is the rubric
Removed experiment: spec-style contract	Slimmed <code>TASK_CONTRACT.md</code> to what a real ticket says, meet WCAG 2.1 AA from the keyboard without changing the design. The detailed requirements became hidden, discoverable only by testing	The archived runs on both sides of the change, <code>sonnet-bench1</code> saturated at 12/12 under the spec-style contract and <code>sonnet-bench2</code> ran against the realistic one, both in <code>runs-archive/</code>	Handing the model a complete spec of a small page removes any need for iteration or verification. Real audits never come with the rubric attached, so neither should the eval
Baseline, benchmark v2	Same one-shot call against the slimmed contract	<code>runs/baseline-2026-08-30T15-56-30</code> , 11/12 pass, \$0.62 total. The miss is 11-lang-title, the easiest case, one axe violation left	Hiding the rubric barely moved the one-shot result, and the single miss is a stochastic slip on a trivial case, not a discovery failure. Sonnet 5 one-shots small textbook pages regardless. A one-case gap at one sample is noise, so pass at first attempt cannot be the headline metric on these fixtures

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Iteration 1	<code>npm run agent -- --variant v1-notools</code> on benchmark v2, iteration but no verification tools	<code>runs/agent-v1-notools-2026-08-30T16-11-14</code> , 12/12 pass, \$1.53 total, mean 8.4 turns	Iteration alone recovers the slip but the margin is within noise. Motivates two follow-ups: measure reliability across repeated sweeps instead of single-shot pass rate, and rerun the ladder on a weaker model where scaffolding has room to matter
Reliability ladder, haiku baseline x3	Ran the one-shot baseline three times on claude-haiku-4-5 to measure the weak-model floor and run-to-run variance, since sonnet saturated the fixtures either way	<code>runs-archive/baseline-haiku-1</code> through <code>-3</code> , 4/12, 7/12, 5/12, about \$0.09 per sweep. Cases 02, 07, 08, 09, 10 never pass, only four cases pass every sweep	The benchmark discriminates sharply once raw model knowledge stops carrying it. One-shot quality is close to a coin flip per case, and worst-of-three is 4/12
Reliability ladder, haiku v3-verify x3	Same model plus the audit, keyboard and screenshot tools, three sweeps	<code>runs-archive/agent-haiku-1</code> through <code>-3</code> , 11/12, 12/12, 11/12 at about \$0.75 per sweep. Eleven cases pass every sweep. Both misses are 07-modal, one genuine (focus return missed), one voided by API-stall timeouts, recorded as failReason timeout in the archived results	Verification tools move the reliability floor from 4/12 to 11/12 and concentrate their gains exactly on the cases axe cannot see. Focus management is the hardest defect class in the set
Infrastructure interlude	Two sweep attempts were voided by API-side request stalls. Hardened the runner with a 20 minute fuse, one automatic fresh-copy retry per case, and strictMcpConfig to keep account connectors out of sessions	The attempts and failReason fields inside <code>runs-archive/agent-haiku-3</code> runner records	Long agent evals need the same reliability engineering as the agents themselves. A timeout must be attributed to weather or capability before it becomes a number

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Tool ablations, haiku, one sweep per rung	Ran the intermediate capability rungs to locate where the gain enters	<code>runs-archive/ablation-v1-notools</code> 9/12, <code>ablation-v2-audit</code> 10/12, <code>ablation-v4-skill</code> 12/12, <code>ablation-v5-memory</code> 11/12, alongside the three-sweep v3 band of 11 to 12	The gain dissociates cleanly. Iteration alone reaches 9/12, the audit tool recovers every scanner-visible case (10/12, leaving only the modal and dropdown, both axe-clean), and the keyboard tool closes those behavioral cases. The skill's sweep was the first perfect 12/12, steadying the modal's focus return. Memory was neutral at one sample, 11/12 with a dropdown flake, and its notes file shows real accumulated lessons. Intermediate rungs are single samples, the endpoints carry the multi-sweep evidence
Final	Promoted every evidence run into <code>runs-archive/</code> , fifteen including the untreated floor, regenerated the aggregated report, verified key-free	<code>runs-archive/REPORT.md</code>	Main contribution: a browser-level verification harness and matching agent tools that let a cheap model do reliably what it cannot do alone, measured as worst-of-N rather than pass at first attempt

WORKING NOTES

- 2026-08-30: benchmark v1 saturated by both arms, contract slimmed, benchmark v2 defined.
- 2026-08-30 to 31: haiku reliability ladder run, two attempts lost to API weather, runner hardened, third sample landed clean.
- 2026-08-31: tool ablations completed, the rung table above locates the gain. Scanner tool for scanner-visible defects, keyboard tool for behavioral ones, skill for the subtlest focus management.